Practical 16: Static and Dynamic Broadcast Receivers

Source: MODULE 2.pdf | Source pages: 50–54

Aim

To develop and test an Android application demonstrating static and dynamic broadcast receivers.

Steps to Perform the Practical

1. Create an Android Studio project for the broadcast receiver practical.

2. Follow the supplied layout and Java code for the receiver and broadcast action.

3. Create the receiver class and implement onReceive() to handle the broadcast.

4. Register the receiver statically in AndroidManifest.xml or dynamically in the activity as shown in the source practical.

5. Run the app and trigger the broadcast action used by the code.

6. Verify that the receiver responds as described in the practical. Note that Android versions may restrict some manifest-declared broadcasts.

Result

The application is run and the expected behavior is verified according to the practical instructions and code.

Note: Use the package name and resource names from the source code consistently. If Android Studio reports a missing resource or class, check spelling and ensure all files are in the correct folders.

9. Develop an Android application implementing Static and Dynamic Broadcast Receivers.

Project Name: BroadcastDemo

File Name: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:gravity="center">

   <TextView

       android:layout_width="wrap_content"

       android:layout_height="wrap_content"

       android:text="Broadcast Receiver Example"

       android:textSize="22sp"/>

</LinearLayout>
```

Create a Broadcast Receiver Class.

Right Click:-> java package → New → Java Class Class Name: MyReceiver

```
package com.example.broadcastdemo;

import android.content.BroadcastReceiver;

import android.content.Context;

import android.content.Intent;

import android.widget.Toast;


public class MyReceiver extends BroadcastReceiver {


   @Override

   public void onReceive(Context context, Intent intent) {

       Toast.makeText(

               context,

               "Broadcast Received",

               Toast.LENGTH_SHORT

       ).show();

   }

}
```

Static Broadcast Receiver.

Open:AndroidManifest.xml

Inside: <application>

Add:

```
<receiver
   android:name=".MyReceiver"
   android:exported="true">
   <intent-filter>
       <action android:name="android.intent.action.AIRPLANE_MODE" />
   </intent-filter>
</receiver>
```

```
        android:supportsRtl="true"
        android:theme="@style/Theme.BroadcastDemo">

        <receiver
            android:name=".MyReceiver"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.AIRPLANE_MODE" />
            </intent-filter>
        </receiver>

        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
```

Dynamic Broadcast Receiver.

File Name: MainActivity.java

```
package com.example.broadcastdemo;

import android.content.IntentFilter;

import android.os.Bundle;

import androidx.appcompat.app.AppCompatActivity;

import androidx.core.content.ContextCompat;


public class MainActivity extends AppCompatActivity {


   MyReceiver myReceiver = new MyReceiver();


   @Override

   protected void onCreate(Bundle savedInstanceState) {

       super.onCreate(savedInstanceState);

       setContentView(R.layout.activity_main);


       // Dynamic Receiver

       IntentFilter filter = new IntentFilter();

       filter.addAction("android.intent.action.AIRPLANE_MODE");


       ContextCompat.registerReceiver(

               this,

               myReceiver,

               filter,

               ContextCompat.RECEIVER_EXPORTED
```

```
        );
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        unregisterReceiver(myReceiver);
    }
}
```

Turn on airplane mode

